All solutions to graded problems must be typed in LaTeX. All other problems can be typed or written, but must be completed. Graded problems must be organized in numerical order. All problem sets (both graded and non-graded) must be uploaded to Canvas as a single PDF document. Students can find the grading scale in our Syllabus.

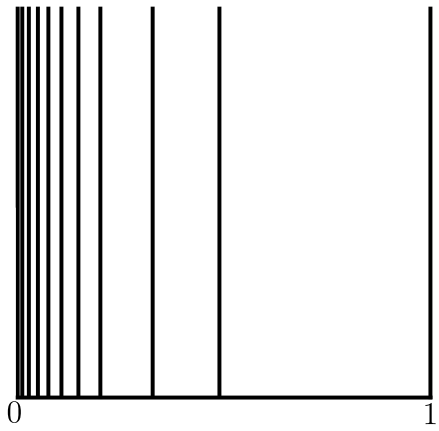
Graded

Exercise 1. Let $\mathscr{T}$ and $\mathscr{T}'$ be topologies on X. Suppose $\mathscr{T} \subset \mathscr{T}'$. What does connectedness of one topology on X imply about connectedness for the other topology?

Exercise 2. What are the connected components of $\mathbb{R}_l$, the real line with the lower limit topology? What are the continuous functions $f : \mathbb{R} \to \mathbb{R}_l$?

Exercise 3. Let $K = \{\frac{1}{n} : n \in \mathbb{Z}_+\} \subset \mathbb{R}$. Define the subspace B of $\mathbb{R}^2$ by

$$B = \big(\{0\} \times [0,1]\big) \cup \big(K \times [0,1]\big) \cup \big([0,1] \times \{0\}\big).$$



Prove that B is path connected but not locally path connected.

Non-Graded

Exercise 4. Let X be a topological space. A point $p \in X$ is called a *cut-point* if $X - \{p\}$ is disconnected. Prove that the property of having a cut point is preserved under homeomorphisms.

Exercise 5. A space is *totally disconnected* if the only connected subspaces are one-point sets. Show that if X has the discrete topology, then X is totally disconnected. Is the converse true?